

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Cord prolapse

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Definitions

Umbilical cord prolapse is defined as descent of the umbilical cord through the cervix alongside or below the presenting part.

it is either:

1. Occult cord prolapse:

Where the cord lies adjacent to the presenting part.

2. Overt cord prolapse:

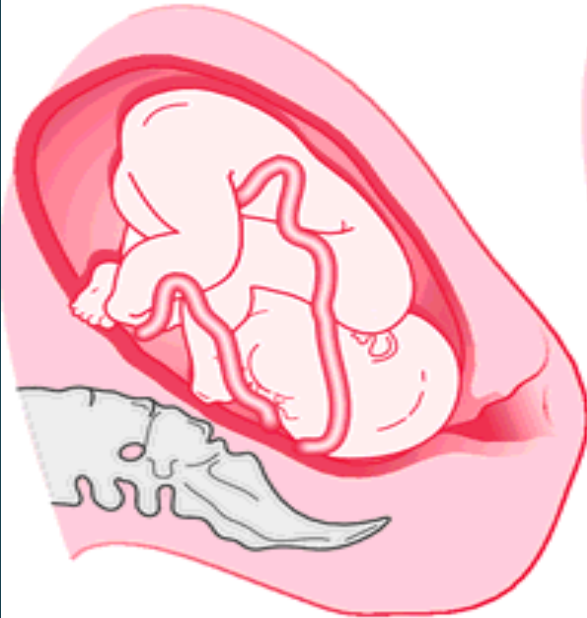
Where the cord lies below the presenting part and the membranes are ruptured.

3. Cord presentation:

This mean descent of the umbilical cord below the presenting part but the membranes are still intact.

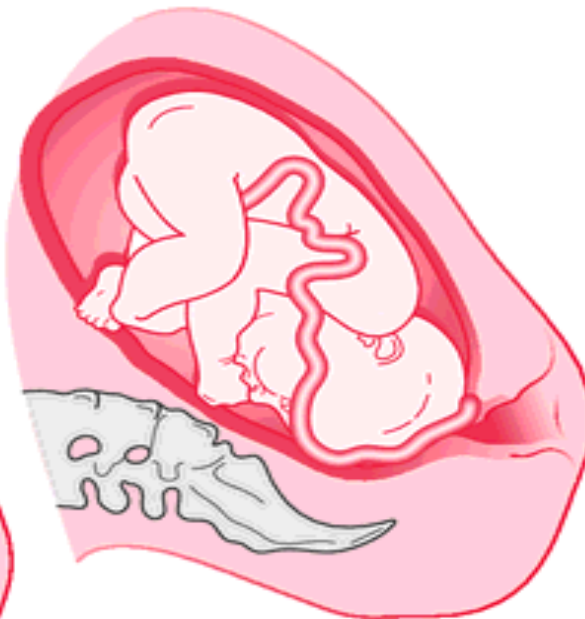
Types of cord prolapse

Occult (hidden) prolapse



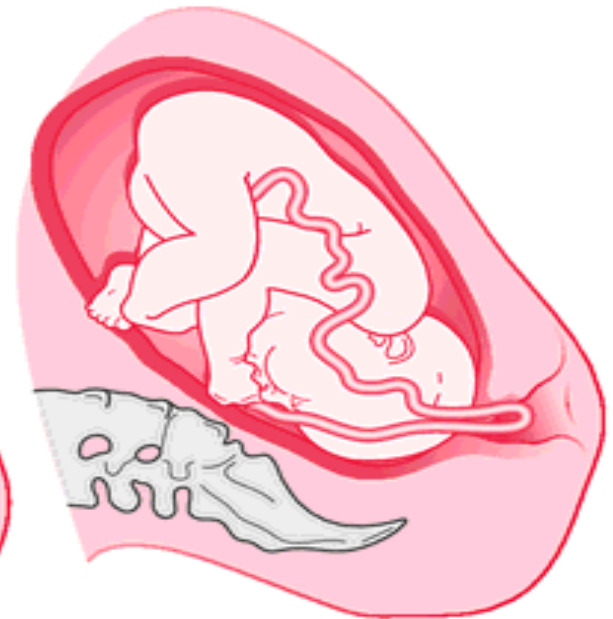
The cord is compressed between the fetal presenting part and pelvis but cannot be seen or felt during vaginal examination.

Cord prolapsed in front of the fetal head



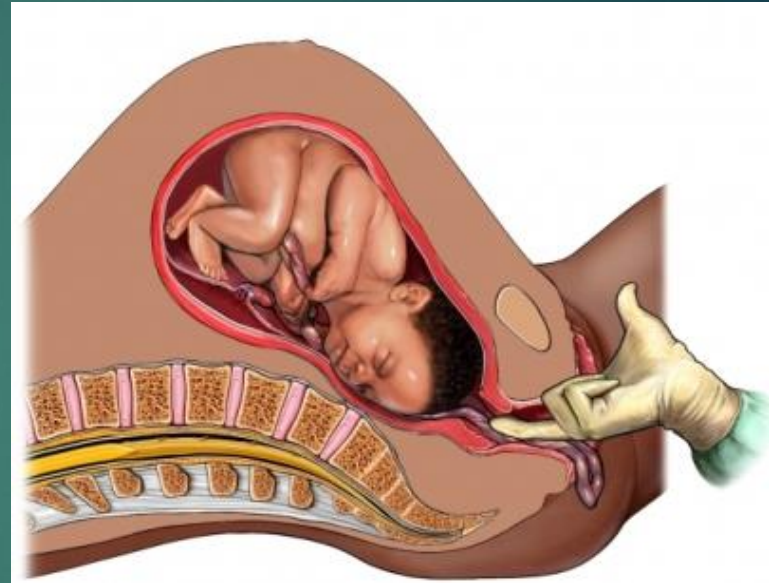
The cord cannot be seen but can probably be felt as a pulsating mass during vaginal examination.

Complete cord prolapse



The cord can be seen protruding from the vagina.

Complete or overt prolapse



Incidence

- It is estimated to occur in 0.1–0.6% of pregnancies with the perinatal mortality rate estimated at 91 per 1,000.
- The incidence of overt cord prolapse depend on the type of presentation of the fetus;
 - For cephalic presentation it is 0.5%
 - For breech presentation depend on the type of breech.
 - Frank breech it is 0.5%
 - Complete breech 5%.
 - Footling breech 10%.
 - Transverse lie 20%
- The incidence of occult prolapse
This is unknown because it can be detected only by fetal heart changes characteristics of umbilical cord compression..

Risks of cord prolapse:

- Prolapse of the cord will expose the cord to intermittent compression between the presenting part and the pelvic inlet, cervix and vaginal canal.
- This compression of the umbilical cord will compromise the fetal circulation and depending on the duration and intensity of compression may lead to fetal hypoxia, brain damage and death.

This will increase the perinatal morbidity and mortality which approach 25% for overt prolapse

Causes:

Any obstetric condition that predisposes to poor application of the fetal presenting part to the cervix may result in prolapse of the umbilical cord.

- 1. Prematurity**
- 2. Multiparity.**
- 3. Abnormal presentation and lie (breech, face, brow, Compound, and transverse and oblique lie .**
- 4. Polyhydramnios and multiple gestation.**
- 5. Placenta previa.**
- 6. Cephalopelvic disproportion.**
- 7. Premature rupture of the membranes before engagement of the head,**

Diagnosis:

-Overt cord prolapse may be diagnosed simply by palpating loops of the cord in the vaginal canal during vaginal examination.

Cord presentation is also diagnosed by feeling loops of cord through the membranes.

Occult prolapse is rarely palpated during pelvic examination. The condition is suspected if fetal heart shows changes which indicate cord compression (e.g. variable deceleration, bradycardia or both).

Management:

Overt cord prolapse:

Once the diagnosis of cord prolapse made an immediate actions done to preserve the life of the fetus (to deliver the fetus as rapidly as possible).

-Vaginal examination done to assess the:

- Degree of cervical effacement and dilatation.
- Station of the presenting part.
- Strength and frequency of pulsation within the cord vessels.

Management:

- When diagnosed, summon senior help and prepare operating theatre for emergency delivery.
- Attempt to prevent further cord compression by elevating the presenting part by:
 - Placing patient in the knee- elbow position, or left lateral Trendelenburg position.
- The examiner should apply continuous upward pressure against the presenting part to lift and maintain the fetus away from the prolapsed cord until preparations for C/S are complete.
- Oxygen (100%) should be given to the mother.
- C/S should be done as rapidly as possible.
- A pediatric team should be standing by for immediate resuscitation of the new born.

Another method to relieve pressure on the cord

Bladder filling:

- This is a method of relieving pressure on the umbilical cord.
- It will raises the presenting part of the fetus off the compressed cord for an extended period of time.
- A number 16 Foley catheter is placed in the urinary bladder.
- The bladder is filled with 500 ml normal saline.
- The balloon is then inflated , the catheter clamped and the drainage tube and urine bag are attached and secured.
- Then the bladder is emptied by unclamping the catheter before opening the peritoneal cavity for C/S

Vaginal delivery may be attempted in two conditions only:

- 1.If the cervix is fully dilated with no Cephalo-pelvic
disproportion delivery can be completed with vacuum
extractor.
2. If the fetus is immature or dead.

Occult cord prolapse:

- If cord compression pattern (variable deceleration) of the fetal heart are recognized during labour:
 - An immediate pelvic examination done to rule out overt cord prolapse.
- The patient placed in lateral Sims or Trendelenburg position to alleviate cord compression.
- If fetal heart return to normal labour allowed to continue with continuous fetal heart monitoring.
- If the cord compression pattern persist or recurs (moderate to severe variable decelerations or bradycardia) rapid C/S done.
- Patient with cord presentation at term should be delivered by C/S

Prognosis:

- Fetal mortality and morbidity rates are high.
- The prognosis depends on the prolapse- delivery interval.
- If complete cord compression persist > 5 min. fetal death or damage is inevitable.



Thank you